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RESEARCH ARTICLE

# Detection of Diabetic Retinopathy Using a Multi-Decision Inception-ResNet-Blended Hybrid Model

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**ABSTRACT** Diabetic retinopathy (DR) is a severe complication of diabetes that affects the retinal structures and can lead to significant visual impairment or even blindness. Early diagnosis is crucial for reducing and preventing the progression of this condition. However, detecting DR's early stages remains challenging due to subtle symptoms that are difficult to recognize independently. Our proposed model leverages 172 weighted layers to analyze both sequential and non-sequential fundus images for effective DR detection. By incorporating a multi-layered transfer learning approach, 86 layers are used for processing color fundus images, while the remaining 86 layers focus on grayscale images. The model undergoes thorough pre-processing and testing phases, utilizing eight layers of convolutions at each stage to handle various data matrices and integrate global and specialized features. The chi-square testing mechanism refines the evaluation of test cases, contributing to the model's overall performance. Using multi-decision hybrid techniques, the model achieves a detection accuracy of 98.1%, outperforming other existing models.

**INDEX TERMS** Adam optimization, clustered class, convolutional neural network, deep learning, dual image, diabetic retinopathy, mild DR.

## I. INTRODUCTION

Diabetes is a disease characterized by continuing metabolic syndrome, and it disrupts the ability of the body to supply the hormone insulin. The hallmark of type 1 diabetes, sometimes referred to as insulin-dependent diabetes, is the immune system's destruction of the pancreatic cells that produce insulin. Type 2 diabetes is a chronic illness in which insufficient insulin synthesis and insulin resistance lead to elevated blood sugar levels. Type 1 diabetes is insulin-dependent diabetes, and this reaction destroys the cells of the pancreas that generate insulin [1], [2]. Figure 1 shows that the process can develop months or years before symptoms occur. If the blood glucose level is too high and the body is not producing, the hormone called insulin is called type 1. Type 2 is a long-term condition in which high levels of sugar circulate in the bloodstream, leading to heavy blood sugar levels. Heavy blood sugar levels are associated with circulatory, immune system, and nervous system disorders [3]. The common causes of type 2 diabetes are overweight, obesity, and an inactive lifestyle. The dataset was collected from patients in the clinic of Sylhet, Bangladesh, via straight surveys and analysis. Despite the occurrence of comparatively extensive symptomlessness, early detection of diabetes is continuously anticipated for a clinically expressive outcome [4]. Almost 50% of people who have diabetes

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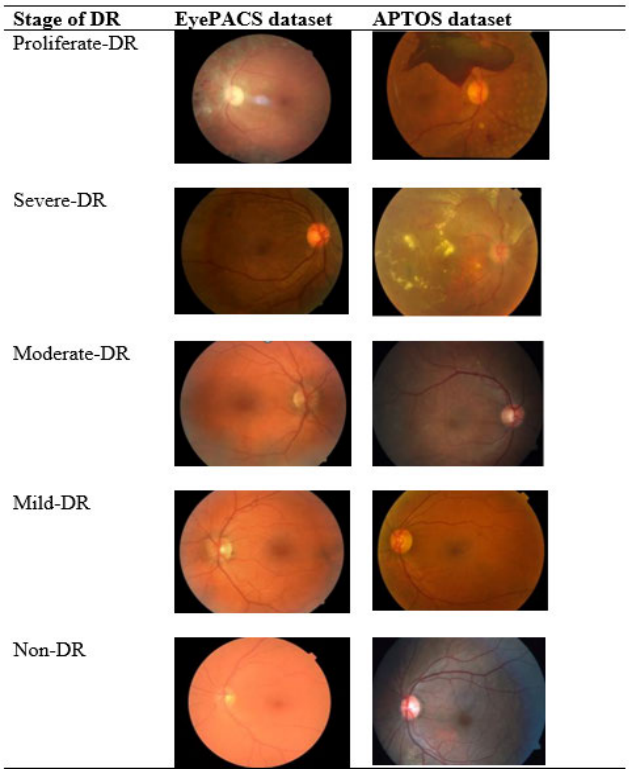



**FIGURE 1.** Diabetic retinopathy (DR) images of 5-stage occurrence samples from the EyePACS and APTOS datasets.

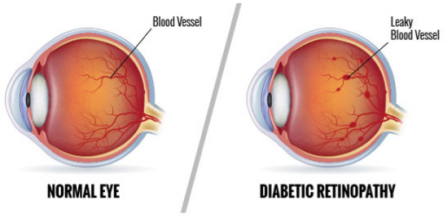



**FIGURE 2.** Sample test image for recognition of inconsistency among regular eye blood vessels and diabetic retinopathy leaking blood vessels.

are undiagnosed. The diabetic retinopathy (DR)-implicated 5-stage occurrence sample images are shown in Fig. 1. These images are collected from the EyePACS and APTOS datasets for the primary cause of test images [5], [6].

Diabetic retinopathy or diabetic eye illness is a collection of eye difficulties that can disturb people with diabetes. These situations include diabetic retinopathy and diabetic macular edema. Diabetes mellitus and hypoglycemia in individuals with polyphagia are caused by low blood sugar levels. Hyperglycemia is caused by elevated blood sugar levels, nervousness, blood pressure, and binge eating disorders. Hyperthyroidism is characterized by increased levels of thyroid hormones and the development of premenstrual disorders, such as some psychiatric disorders, medications like corticosteroids, and common medical illnesses, including Prader-Willi and Kleine-Levin syndrome.

This research has three major objectives:

- This study presents a multi-decision inception-ResNet blended hybrid model that can process both sequential and non-sequential images. It does this by integrating many layers of dual image-based parameters.
- A multilayered transfer learning process is used to train the suggested model. It consists of 172 weighted multi-layers displayed with 86 weighted layers linked to color fundus images and 86 additional weighted layers linked to black-and-white photos.
- It was pre-processed with various training phases and supporting sub-testing stages. Every stage has integrated eight layers of convolutions with dependable, non-dependable data matrices, which have incorporated the number of repetitions of internal connections by combining global and specialized features.

The model perform on unseen datasets that differ from the EyePACS and APTOS datasets with best outcome results of continuous evaluation and incorporation of augmentation. Continuous testing on diverse datasets is essential to ascertain the model's generalizability. Regular performance evaluations can help identify potential overfitting and inform adjustments in training strategies. Implementing data augmentation during training can also help mitigate overfitting, as it exposes the model to a wider range of variations, potentially improving its robustness against unseen data.

Evaluating the performance of a model on unseen datasets that differ from the EyePACS and APTOS datasets is crucial for assessing its generalizability and robustness, especially for applications like diabetic retinopathy detection. Here's an analysis of how the model might perform and whether there is evidence of overfitting:

### A. PERFORMANCE ON UNSEEN DATASETS

#### 1) GENERALIZATION ABILITY

If the model trained on EyePACS and APTOS datasets performs well on unseen datasets, it indicates that it has learned to identify relevant features associated with diabetic retinopathy rather than memorizing the training data. The ability to generalize suggests that the model captures the underlying patterns of the disease.

#### 2) VARIABILITY IN DATA

However, if the unseen datasets exhibit significant differences in image quality, acquisition methods, or demographic factors (e.g., age, ethnicity) compared to the training datasets, the model's performance may decline. This decline could be due to the model's reliance on features that are not universally applicable across different populations or imaging conditions.

### B. EVIDENCE OF OVERFITTING

#### 1) TRAINING VERSUS VALIDATION PERFORMANCE

If the model shows high accuracy on the training set but significantly lower accuracy on validation and unseen datasets, this could indicate overfitting. Overfitting occurs when the model learns noise and specific details of the training data rather than generalizable features.

#### 2) LEARNING CURVES ANALYSIS

Examining learning curves can provide insights into overfitting. A significant gap between training and validation accuracy, particularly when the training accuracy continues to improve while validation accuracy plateaus or decreases, suggests overfitting.

#### 3) PERFORMANCE METRICS

Metrics such as precision, recall, F1-score, and area under the ROC curve (AUC) on unseen datasets can help evaluate the model's effectiveness. A notable drop in these metrics compared to training metrics may signal overfitting.

### C. ADAPTABILITY AND ROBUSTNESS

#### 1) ROBUSTNESS TO VARIATIONS

A robust model should maintain performance across datasets with different characteristics. If the model shows resilience to variations in illumination, resolution, or artifacts, it indicates that it has learned more general features, thus mitigating overfitting concerns.

#### 2) TRANSFER LEARNING CONSIDERATION

If the model employs transfer learning techniques, where it adapts a pre-trained model on more diverse datasets, this may enhance its ability to generalize to new, unseen data, further reducing the risk of overfitting.

This article is structured subsequently: Section II encompasses the allied continuous existing works along with their research background. Section III describes the proposed methodology for deep learning (DL) based ResNet-ResNet algorithm sequences with active dataset parameters. Section IV describes the integration of the experimental setup and execution of various disparity parameters that directly and indirectly impact the DR stages. Section V provides the experimental results obtained through multi-decision inception-ResNet via the integration of various transfer learning mechanisms. Section VI addresses the conclusion and future directions.

## II. RELATED WORK

Existing methods can be grouped into two broad categories: traditional machine learning techniques (classification, clustering) and DL architectures (CNNs, hybrid models). The following subsections elaborate on these approaches.

The authors in [7] introduced a step-by-step procedure for the circle cropping algorithm; it refers to all sides of an eye image when it can be rotated in all directions. The directions of the image should be up, top, bottom, corners, and edges. The grayscale image scale provides good-quality images. Image quality can depend on resolution, and image resolution can focus on cropping and resizing the image; resizing or rescaling can be a minimum of the actual image and a maximum of the image. After all the techniques are applied, images of good quality and reasonable resolution can be obtained. The diabetic stages are separated into type 1 and type 2. Diabetes is characterized by increased diabetic retinopathy and non-proliferating diabetic retinopathy. The on-proliferative stage included four stages: regular, mild, moderate, and severe. The threshold values for no diabetic retinopathy were 0, 1, 2, and 3. A threshold value of four is considered to indicate proliferative diabetic retinopathy [8].

The authors of [9] proposed an innovative hybrid model built by integrating a genetic algorithm (GA)-nominal network (NN) with an NN-fuzzy logic control inference system (FLCIS). Neural networks and fuzzy inference systems work autonomously of each other. The rate of diabetic retinopathy in people is likely to deteriorate at a critical speed. The author discussed the early visual defects and processes clinically significant for diabetic retinopathy. Diabetic mellites are divided into two types: type 1 and type 2. This approach used single-field digital monochromatic non-mydriatic photography. In addition, it aims to analyze a physician utilizing an orthoscope with 30 degrees of stereoscopic color [10]. Gardiner et al. [11] added a new indication for multifunctional and microvascular SMCs in diabetic retinopathy. Moreover, neural cells suffer and die due to microvascular partitioning, which causes the phagocytic arrangement of the JVM. The authors in [12] proposed a method that uses tenfold cross-validation to provide the best accuracy; the significance of these efforts is that of novel analytic techniques. For more information on 3D-OCT, see the OCT CAD classification. The author mainly focused on the early detection of diabetic retinopathy; in this study, there were two types, type 1 and type 2, which can be non-proliferate or proliferate, respectively. Non-proliferative disease can be classified as mild, moderate, severe, or proliferative. The proposed method can automatically estimate 3D-MGRF model-based spatial occurrence. The authors in [13] proposed methods built on segmentation by a unit algorithm, and the similarities between blood vessels, retinal hemorrhage, contextual work of original images, and the segmentation of images with digital fund images were described. The image can be divided into several segments for comparison of diabetic retinopathy; if blood vessels have any leakages of background based on segmentation, we can identify them, and vision loss can be avoided. DR is a disorder of diabetes that creates uncertainties through various occurrences in human vision. It occurs due to harm to blood vessels based on light-sensitive tissues in the eye retina [14].

DR is sourced by the microvascular obstacle of diabetes, which manifests as morphological transforms in the eye fundus. The risk of vision loss is reduced if the DR is largely identified at its early stages, which can aid in diagnosing and treating DR. DR can be determined noninvasively at its early stages by developing and integrating color fundus photography [15]. The CNN-based DL approach aims to define data structures and widely construct spatial orientations and interpretations to construct models [16]. The authors of [3] developed a CNN model with ten layers to identify microaneurysm lesions. The authors in [4] proposed a methodology of the CNN-Patch technique and loss function by integrating

the E-Ophtha, ROC, and DIARETDB1 datasets by holding the free-response receiver operating characteristic curves. The analysed literature has proposed articles and built models for classifying and detecting DR stages. Some research has been conducted based on image classification-related tasks, which have performed and clustered the techniques used for conventional image processing along with supporting matrices [17]. The authors proposed and discussed the detection and segmentation of DR based on retinal images implicated in blood vessels [18]. The authors present and discuss methods for detecting colour fundus images based on exudation [19]. Some research has been performed based on image classification-related tasks with the implications of recent DL procedures [20].

The authors presented and discussed profound learning-based classification advances tuned based on DR binary classification, which is nonreferable and referable [21]. The authors in [22] proposed DR-based binary and multi-class classification, accomplished using Haralick and multi-resolution features. The authors described the DR severity stages of consequences by using classification models [23]. The authors in [24] proposed a hybrid model by integrating five pre-trained CNN models, Dense121, Inceptionv3, ResNet50, Dense169, and Xception, which are used for DR classification in five stages of DR occurrence. The authors in [25] integrated the Kaggle EyePACS dataset for the classification of DR stages and their occurrences. They described the comparative analysis of GoogleNet, AleXNet, and ResNet with VggNet in terms of the best accuracy outcomes. The authors in [26] proposed a deep convolution neural network that detects DR in the initial two stages, such as normal DR and NPDR. Another study investigated automated image processing approaches integrated with microaneurysm case-based detection [27].

The model was evaluated by an Xception transfer learning model, which helped train a large amount of data. This proposed model has been integrated and evaluated with two bulky public-domain Kaggle datasets by APTOS [28] and EyePACS [29]. The authors of [30] proposed a hybrid model by integrating various DL models, which are used to discover different DR stages, such as no DR, referable DR (rDR), and vision-threatening DR (vtDR), with macular edema. The authors of [31] proposed the Bossa-Nova and Fisher vector midlevel feature-based approach for the classification of DR. This approach integrates features to extend visual word feature-based classical Bags. Some research has been performed based on image classification-related tasks with the implications of recent DL procedures. The authors in [32] suggested a convolution neural network that was individually used and applied to right and left eye images by using DR binary classification. The authors of [33] integrated the EyePACS dataset with three DL models—a feed-forward neural network, a deep neural network, and a convolutional neural network—for classifying the DR stage and computed and analyzed the performances of different CNN models on the EyePACS dataset.

In contrast, concurrent segmentation of red and bright lesions in fundus images uses a multitasking architecture [34]. The authors of [35] proposed a region-specific multitask detection model to classify thirty-six different retinal diseases without exploratory DR classification stages. The authors of [36] proposed an algorithm that focused mainly on age and determined how people are affected by diabetes; these two types of diabetes have been identified. Nevertheless, the author's study says that both types are the same, but they alert people about their food habits and glucose levels. The authors in [37] and [38] presented DL-based CNN techniques, which have integrated end-to-end learning capabilities that are unnecessary for formulating handcrafted features to generate the best outcomes in the field of image classification and segmentation. The authors in [39] proposed a DR classification method based on binary-type implicated 3-CNN models. Some research has been conducted based on image classification-related tasks, which have performed clustered techniques for conventional image processing and supporting matrices [40], [41], [42].

Most models have evaluated and analyzed the detection and classification of DR color fundus images using DL techniques [44]. A 2D convolution neural network model is used in another study to determine the author's effective prototype approach [45], which has an accuracy of 99% across the representation of 10 epochs for the training and testing sets. A deep neural network (NN) and genetic algorithm (GN) feature selection method were presented in another study [46]. Features are extracted from fundus images using five advanced CNN architectures. A genetic algorithm is then used to choose features and rank them into optimal and unsatisfactory ranks. Next, the training, along with validation feature vectors' non-optimal feature characteristics, is removed. The DR recognition model was developed using a support vector machine (SVM) classification approach. The ensemble feature vector was created by combining local binary patterns with DL features, later which will be optimized using the algorithmic sequences of binary dragonfly and the sine cosine [47]. A different research model that was suggested used an image pre-processing technique to overcome the noise and artifacts seen in fundus images. It was suggested to use the You Only Look Once V7 technique for a feature extraction process [48]. In addition to integrating percentage ratios of 80:20 for the best and poorest quality images in training and testing, the author suggested the robotic decision-making ResNet feed-forward NN strategy for DR detection [48]. The 10-fold cross-validation technique was also mentioned [49]. The test instances that are normal or abnormal are covered in a different study [50]. Principal component analysis is combined with DL for dimensionality reduction. It uses the Harris Hawks optimization practices for additional optimization of the feature selection, mining, and classification process to extract the largely significant features [51]. An AWS web app based on machine learning was suggested by another study [52].

The DL approaches related to drug discovery have been applied in real-world test case settings [53]. The retrieved features exhibit improved spatial and channel awareness. The network model's performance is optimized by adjusting the loss function, initial parameters, and small-step learning approach [54]. In another study, the retinal images were subjected to an enhanced pooling function and an activation function in the ResNet-50 model for autonomous lesion identification with less loss and processing time [55].

## III. METHODOLOGY

The convolutional neural network (CNN) model learns the desirable features and matrices used to mine it for medical image processing and analysis purposes [43]. CNN is based on convolution layers that receive input and then transform the data from the pre-processed image and exceed it as input values to the next processed layer. Every conversion is treated as a convolution-based operation. The CNN architecture is framed with a convolutional filter representing the crucial function of a self-learning mechanism for extracting the necessary features for proficient medical image processing and analysis. Each convolution layer is framed with several filters to sense patterns such as shapes, edges, curves, objects, colors, textures, and small colored patches. The image kernels are filtered in the form of $2 \times 2$, $3 \times 3$, or $4 \times 4$ vector ratios, which have a small-large matrix functional to an intact image. The convolution layers are pooled together to obtain the sample input representing the input image, and its dimensionality is reduced by preserving the activated features of utmost value in the sub-region binding mechanism. The stride functionality describes the movement of the surrounding number of image pixels of the input. It automatically senses the surrounding pixels to move from one to another by shifting the filter to 1 pixel at a time if the stride is 1—and the filter to 2 pixels at a time if the stride is 2. Finally, the filter was shifted to 3 pixels at a time if the stride values were three, and the same process continued for further sliding values. The size of a large image will be reduced to a moderate size with the implications of stride values along with correlated filters.

This research adopts DL-based CNNs to accomplish several goals. An exploratory research study will conduct a comprehensive investigation into AD. The main goal of my research is the second objective, whereby we present a novel framework and utilize it for the public dataset. Due to DL's ability to meet both supervised and unsupervised segmentation, the technique module that is being proposed could satisfy unlabeled data when training the picture with labeled deep understanding. An empirical evaluation will be carried out to evaluate the feasibility of the proposed framework. This research has suggested a DL-based dual-image multilayer ResNet algorithm sequence approach, which mainly focuses on detecting and identifying early-stage DR, such as non-DR, mild DR (MiDR), moderate DR (MoDR), severe DR (SeDR), and proliferative DR (PrDR). All dependable DR stages are classified and analyzed based on sequential and non-sequential images using automatic image-mapping techniques.

### A. MODEL ARCHITECTURE

The algorithmic sequences of the DL Original-ResNet defined as

**Step 1:** Diabetes symptoms classified: partial paresis, baldness, age, sex, weakness class, genital thrush, polyuria, visual blurring, polyphagia, delayed healing, abrupt weight loss, itching, irritability, and obesity degree.

**Step 2:** Class characterize the positive, true positive (TP), true negative (TN), false positive (FP), false negative (FN) values of diabetic symptoms.

**Step 3:** Data preparation and collection

cim1 is the first instant of the exponential decomposition rate in (AOpt) based on color images.

cim2 is the second moment of the exponential decomposition rate in (AOpt) based on color images.

bwim1 is the initial instant of the exponential decomposition rate in (AOpt) based on black-and-white images.

bwim2: The second instant of the exponential decomposition rate based on black-and-white images in (AOpt)

**Step 4:** Hyperparameters and their minimum and maximum values.

**Step 5:** n-estimators show the number of boosting stages, and the default value is 100.

**Step 6:** Image Type, Conversion and Labelling

Fundus image types:

Fundus images based on color images $\leftarrow$ FCI = fci{fci1, fci2, fci3.... fcin}

Fundus images based on black-white images $\leftarrow$ FBW = fbw{fbw1, fbw2, fbw3.... fbwn}

fundus images p$\in$P; P = {p/p$\in${color$\leftarrow$1, black-white$\leftarrow$0}}

Labeling grades: positive (1), TP $\leftarrow$11, TN $\leftarrow$ 10, FP $\leftarrow$ 01, FN$\leftarrow$00 values

Dual Labelling Mechanism (P, Q) $\Leftrightarrow$ ($\sim$P, $\sim$Q)

where Q1 = {q1$\in${000, 001, 010, 011, 100}} and Q2 = {q2$\in${00, 01, 10, 11}}

Q1 $\leftarrow$ labeling primary case, and

Q2 $\leftarrow$ labeling secondary case based on positive (1), TP (11), TN (10), FP (01), and FN (00) results.

DR consequence$\Leftrightarrow$Inference level (IL) $\Leftrightarrow$ {main-level1.sub-level, main-level2.sub-level, main-level3.sub-level}

Normal $\leftarrow$ 000.00

Mild DR 001 $\leftarrow$ IL1 $\Leftrightarrow${001.01, 001.10, 001.11}

Moderate DR 010 $\leftarrow$ IL2 $\Leftrightarrow$ {001.01, 001.10, 001.11}

Severe DR 011 $\leftarrow$ IL3 $\Leftrightarrow$ {011.01, 011.10, 011.11}

Proliferate DR 100 $\leftarrow$ IL4 $\Leftrightarrow$ {100.01, 100.10, 100.11}

**Step 7:** The step size compresses the subscription of every tree by their value and default value=0.1, which is the trade-off between the step size and n_estimators.

**Step 8:** Max_depth shows the extreme depth of the discrete regression evaluation, where the default value is 3, which is

the maximum depth restriction on the number of nodes in the tree.

Set Activation Function ← Act ← Rectified Linear Unit (ReLU)

Initial plotting value (M)← 0.5

Add Conv Layer (A-CFI-CNL1)← (K, A-CFI, PADD, S, Act) at filter F ← 64

Add Conv Layer (A-CFI-CNL1)← (K, A-BWI, PADD, S, Act) at filter F ← 64

Add Conv Layer (A-CFI-CNL2) ← (K, PADD, Act), L2=10-3, at filter F ← 32

Add Conv Layer (A-CFI-CNL2)← (K, PADD, Act), L2=10-3, at filter F ← 64

Add Conv Layer (A-CFI-CNL3) ← (K, PADD, Act), L2=10-3, at filter F ← 24

Add Conv Layer (A-BWI-CNL3) ← (K, PADD, Act), L2=10-3, at filter F ← 24

Add Conv Layer (A-CFI-CNL3) ← (K, PADD, Act), L2=10-3, at filter F ← 12

Add Conv Layer (A-BWI-CNL3) ← (K, PADD, Act), L2=10-3, at filter F ← 12

**Step 9:** Input → Expected output mechanism

Original input:

Fundus images p∈P; P = {p/p∈{color←1, black-white←0}}

Expected output:

The trained model forecast possibility scores equivalent to∀q for an input p based on p∈{color←1, black-white←0}.

**Step 10:** Preprocessing Image Techniques using a Dual-Image Multilayer Mapping Approach

Resize the image to FCI = fci {fci1, fci2, fci3.... fcin} → 312 × 312×3

Resize the image to FBW = fbw {fbw1, fbw2, fbw3.... fbwn} → 312 × 312×3

Perform Data Augmentation techniques:

Zooming →z

Trimming →t

Rotary motion →rm

Horizontal spin →hs

Vertical spin → vs

Breadth move→ bm

Height move→ hm

**Step 11:** Under sampling can define the segment of trials used for correcting the separate base learners, and the default value is 1.0

**Step 12:** Max_features can indicate the number of landscapes to be reflected for the best split; possible values are auto, sqrt, log2, and thedefault value = None.

**Step 13:** DL inception-ResNet Model-based Dual-Image Multilayer Mapping Image-based Classification Model (CM) and Regression Model (RM)

**Step 13.1:** Dual-Image Multilayer Mapping Image-based Classification Model (CM)

Blended Hybrid Model (BHM) ←{{CM1, CM2}, RM} ← momentum (Mt)

Every ∀bhm∈BHM

imlr1 = 0.001, Mt1 = 0.4

imlr2 = 0.005, Mt2 = 0.8

Every imlr = 0.001 to 0.005 and Mt = 0.4 to 0.8

Every epoch=1 to 360, do Every both tiny clusters (Pmini, Qmini) ∈ (P, Q) do

Decision Making (DM) Process-based Test Condition (TC)

Test condition 1: initialized classification model →bhm then

Test condition 2: DM ←bhm, then

Parameters of DM through classification, turning, and integration

Stochastic Gradient Decent optimization (SGD) ← Parameter tuning and integration

TC1: epochs>70

imlr = 0.0001, momentum1 = 0.4

TC2: epochs>140

imlr = 0.0002, momentum1 = 0.5

TC3: epochs>210

imlr = 0.0003, momentum1 = 0.6

TC4: epochs>280

imlr = 0.0004, momentum1 = 07

TC5: epochs>350

imlr = 0.0005, momentum1 = 0.8

**Step 13.2:** Dual-Image Multilayer Mapping Image-based Regression Model (RM)

TC1: initialized regression model →bhm then

TC2: DM ←bhm then

Test condition 3: epochs<70, then

Parameters of DM through classification, turning and integration

Adaptive Moment Estimation (Adam) ← Parameters depending on the integration

**Step 14:** DL inception-ResNet Model-based Dual-Image Multi-Layer Mapping Image feature extraction

Features extracted from ∀bhm∈ BHM and fed to an MLP classifier to generate the dual-image-based multilayer mapping approach.

**Step 14.1:** A dual-image-based multilayer inferences → features extracted from a series of networks

**Step 14.2:** Every dual-image-based multilayer mapping approach

imlr1 = 0.001, Mt1 = 0.4

imlr2 = 0.005, Mt2 = 0.8

Step 14.3:

cim1 =0.7, cim2=0.890

bwim1=0.7, bwim2=0.890

**Step 14.4:**

Every epoch=1 to 70

for each tiny-cluster1 (Pmini, Qmini) ∈ (P, Q) do

Decision Making → dm

Test condition 1: initialized Adaptive Moment Estimation →bhm then

Adaptive Moment Estimation (Adam) ← Parameter turning and integration

**Step 14.5:**

Every epoch=71 to 90

for each tiny-cluster2 (Pmini, Qmini) ∈ (P, Q), do Decision Making → dm

Test condition 2: initialized Adaptive Moment Estimation →bhm then

Adaptive Moment Estimation (Adam) ← Parameter turning and integration

**Step 14.6:**

The parameters of the multitasking model are updated via adaptive moment estimation.

The validation error does not improve for four epochs; thus,

imlr1 = avg ((imlr1 * 0.01) + ((imlr2 * 0.01))

imlr2 = avg ((imlr1 * 0.01) + ((imlr2 * 0.01))

**Step 15:** Building the test cases

**Step 15.1:** for p∈Ptest, do

**Step 15.2:** The trained dual-image-based multilayer mapping approach predicts the probabilityscores for ∀q

fundus images p∈P; P = {p/p∈{color←1, black white←0}}

Labels (P, Q), where Q = {q/q∈{000, 001, 010, 011, 100}}

The proposed DL-based dual-image multilayer ResNet algorithm sequences are framed with 15 sequence steps. The implications of the organization layer data are shown in the table. 1. Step 1 integrates the data values based on the DR impact stages using various symptoms. In step 2, the binary bit values are generated based on five resulting primary parameters, namely, positive, TP, TN, FP, and FN, as shown in Table 2. The computational data were prepared based on the original image and then converted into a colour fundus image as layer-1 and a black-white image as layer-2. The image learning rates were calculated, and two pairs of moments were generated based on the color fundus image and black-white image. Fundus images based on colour images (FCIs) were considered layer 1 (L1), fundus images based on black-white images (FBWs) were considered layer 1 (L2), and the blended hybrid fundus image model (BHM) was considered the blended layer (BL). The FCIs are extended with experimental set values of fci{fci1, fci2, fci3.... fcin}, FBW extended with experimental set values of fbw{fbw1, fbw2, fbw3.... fbwn} and BHM extended with experimental set values of P = {p/p∈{FCI ←1, FBW ←0}}.

The classification and detection of DR stages are integrated using the dual image approach like integration and aggregation color fundus imae and black-white image as shown in Figure 4. Both images are analyzed separately and blended with missing points of each image-sequence of color fundus image and black-white image. The mapping test cases are planned to experiment or execute on sequential and non-sequential images. The 86 layers designated for color fundus images FSC-CFI and the 86 for grayscaleSSC-BWI images contribute to the model's ability to distinguish between different stages of diabetic retinopathy as follows;

DR consequence⇔Inference level (IL) ⇔ {main-level1.sub-level, main-level2.sub-level, main-level3.sub-level}

Normal 000 ← 000.00

Mild DR 001 ← IL1 ⇔{001.01, 001.10, 001.11}

**TABLE 1. Layer-based classified image set values and their labeling sequences.**

| Image Layers | Layer Type | Set Values | Labeling | Resized Image | Classification Model (CM) |
|---|---|---|---|---|---|
| Layer 1 (L1) | Fundus Images based on color images (FCI) | FCI = fci{ fci1, fci2, fci3.... fcin} | Q1 represents the primary case of labelling. Q1 ← q1∈{000, 001, 010, 011, 100} | FCI = fci {fci1, fci2, fci3.... fcin} → 312×312×3 | CM1 |
| Layer 2 (L2) | Fundus Images based on black–white images (FBW) | FBW = fbw{ fbw1, fbw2, fbw3.... fbwn} | Q2 represents the second case of labeling. Q2 ← {q2∈{00, 01, 10, 11} | FBW = fbw {fbw1, fbw2, fbw3.... fbwn} → 312×312×3 | CM2 |
| Blended Layer (BL) | Blended Hybrid Fundus Image Model (BHM) | P = {p/p∈{ FCI ←1, FBW ←0}} | Q1 ← {q1∈{000, 001, 010, 011, 100}} && Q2 ← {q2∈{00, 01, 10, 11}} | [(fci {fci1, fci2, fci3.... fcin} → 312×312×3) && (fbw {fbw1, fbw2, fbw3.... fbwn} → 312×312×3)] - [(fci {fci1, fci2, fci3.... fcin} →312×312×3) ‖ (fbw {fbw1, fbw2, fbw3.... fbwn} → 312×312×3)] | BHM ←{{CM1, CM2}, RM} Where RM← Regression Model |

Moderate DR 010 ← IL2 ⇔ {001.01, 001.10, 001.11}

Severe DR 011 ← IL3 ⇔ {011.01, 011.10, 011.11}

Proliferate DR 100 ← IL4 ⇔ {100.01, 100.10, 100.11}

The first and second moments of the color image are framed based on exponential decomposition rates, which are represented by cim1 and cim2, respectively, in the same way. The first and second moments of black–white are based on exponential decomposition rates, representing bwim1 and bwim1, respectively. Then, the hyperparameters and their minimum and maximum values are integrated into step 5. Critical hyperparameters in the proposed ResNet model that directly affect the network's capacity to learn pertinent features from both color fundus and black-and-white images are the number of filters, kernel sizes, and activation functions. By carefully adjusting these hyperparameters, one can improve the resilience of the model, optimize feature extraction, and eventually improve performance in applications

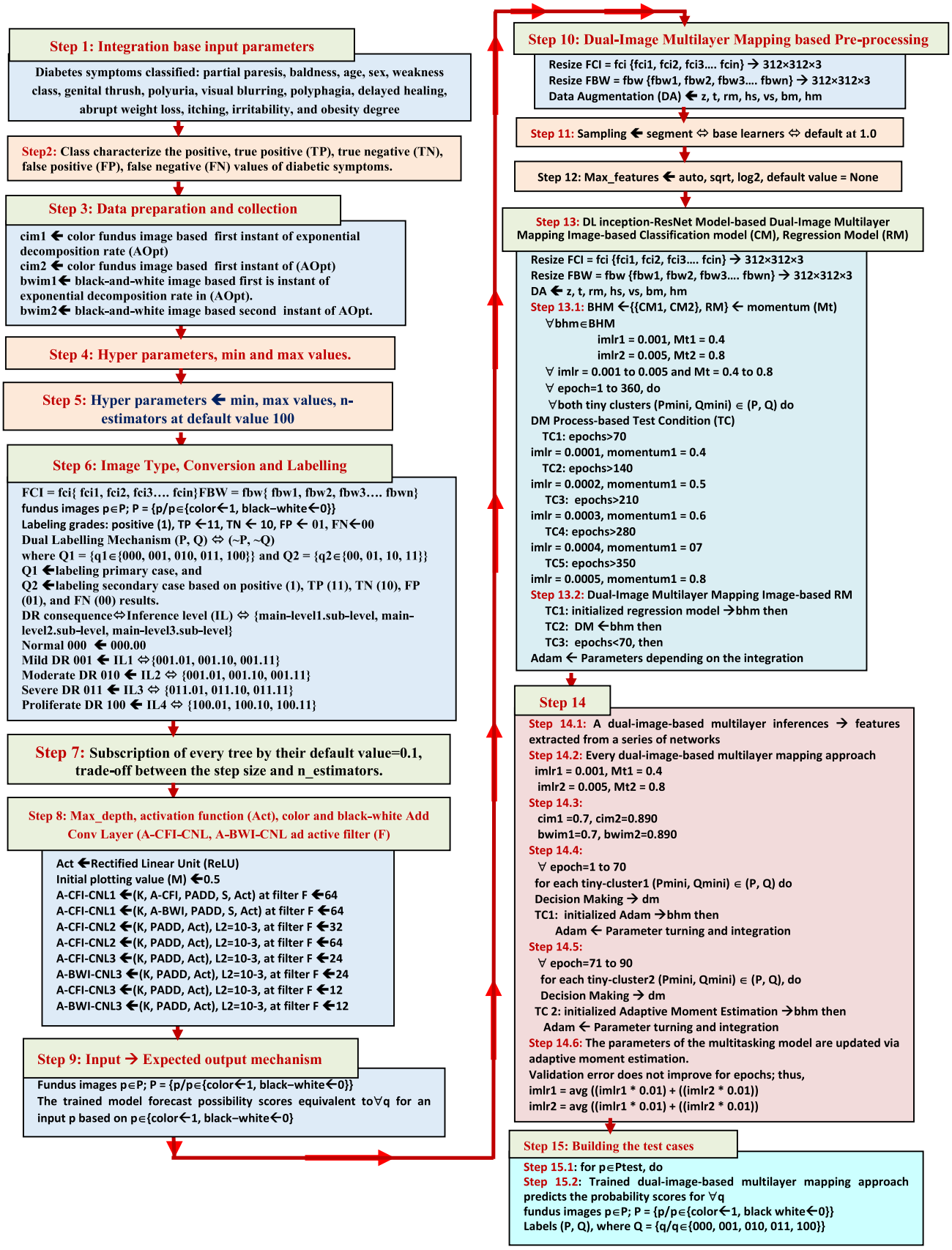



**FIGURE 3.** Step by step process of multi-decision inception-ResNet-blended hybrid model for detection of DR stages.

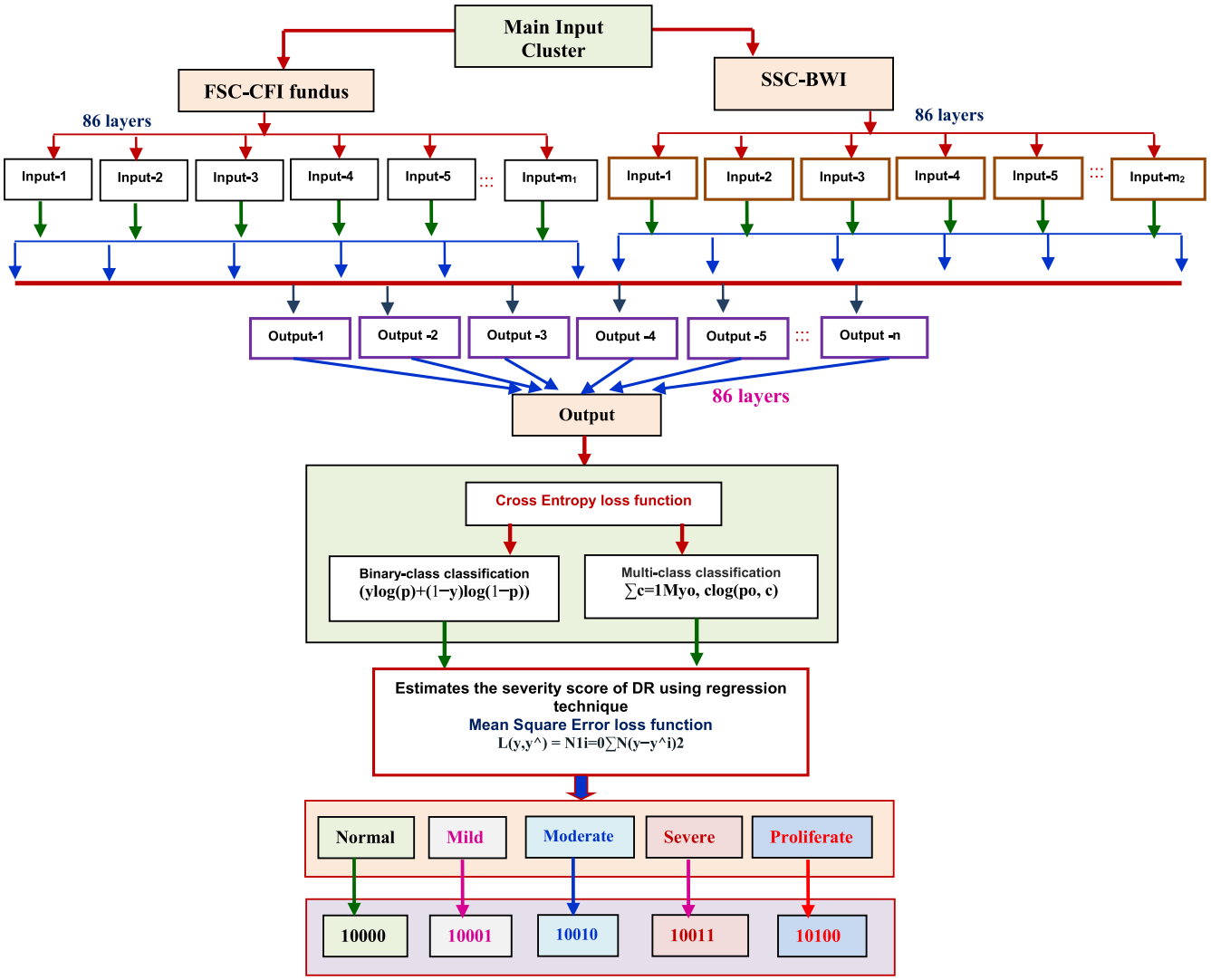



**FIGURE 4.** Classification and recognition of diabetic retinopathy stages using deep learning-based dual-image multi-layer mapping approach DR grades of severity.

like the diagnosis of diabetic retinopathy. Achieving the best outcomes while avoiding problems like overfitting and high computing loads requires balancing these hyperparameters, the network can learn more intricate representations of the data by adding more filters to deeper levels. A careful selection of filter numbers is necessary for color fundus photos since different color channels may expose different features than black-and-white images. Improved performance and quicker convergence in deep network training could be possible with ReLU. Activation functions efficiently control the different intensities across RGB channels in the layer-specific behavior of color fundus images, whereas intensity gradients are the main emphasis of black-and-white images. The ReLU activation function affects how well features are acquired from various kinds of images.

The n-estimators are included for processing the number of boosting stages by keeping the default value=100 as the threshold. In step 6, the labels are labeled and converted according to the image type. The fundus color images are included in one cluster group as FCI = fci {fci1, fci2, fci3.... fcin}, and fundus black-white images are included in another cluster group as FBW = fbw {fbw1, fbw2, fbw3.... fbwn}. Then, the final draft experimental image is integrated as fundus images p∈P; P = {p/p∈{color←1, black-white←0}}. Then, the labeling grades are formulated as positive (1), TP (11), TN (10), FP (01), and FN (00) values. Q1 represents the primary case of labeling, and Q2 represents the second case of labeling based on positive (1), TP (11), TN (10), TP (01), and FP (00) results.

In step 7, the sample size and n-estimators are framed by wrapping the subscription of every single tree node with consideration of their weights by defaulting value=0.1. In step 8, the max_depth values represent the severe depth of the distinct regression estimate with default value = 3. The maximum depth restricts the number of nodes in the tree. The input to the expected output mechanisms is formulated

**TABLE 2.** The various dependable to non-dependable disparity parameters versus causing DR symptoms with a clustering mechanism.

| Dependable to Non Dependable Disparity Parameters (D-NonDDP) | DR Symptoms | Cluster Class 1 (CC-1) | CC-1 Occurrences | Cluster Class 2 (CC-2) | CC-2 Occurrences |
|---|---|---|---|---|---|
| Age | A regular urine capacity depends on a person's age and gender | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| polyuria | Polyuria is the most common caused by the use of diuretics and uncontrolled diabetes mellitus, and urine volume is high | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| polydipsia | Polydipsia symptoms are feeling of extreme thirstiness, and this is especially noticeable when the person can feel the way even after person have drunk a lot of water | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| sudden-weight-loss | Sudden weight loss is a perceptible drop-in body weight that arises uniformly if the person is not attempted to lose weight | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| weakness | Weakness of diabetic retinopathy encompasses symptoms that may include unclear vision, struggle to see colours, and eye spots | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| polyphagia | Polyphagia, diabetic mellitus, and Hypoglycemia are due to low blood sugar levels | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| Genital thrush | Genital thrush disease with specific problematic diabetes because glucose aids candida grows | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| Visual blurring | Visual blurring can occur over time; diabetes harms minor blood vessels throughout the eye as well as the retina | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| Itching | If the person has diabetes, more likely to have dry skin due to glucose. | Diabetes →itchy skin | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| Irritability | Irritability is a feeling of a person; the variety of high and low is not unusual if a person has diabetes | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| Delayed healing | Delayed healing when the person has diabetes | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| partial-paresis | A partial paresis person can struggle to concentrate or have a dual vision | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| Muscle stiffness | Muscle stiffness Symptoms of myalgia, joint pain to move a person's joints | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |

**TABLE 2.** *(Continued.)* **The various dependable to non-dependable disparity parameters versus causing DR symptoms with a clustering mechanism.**

| | | | | | |
|---|---|---|---|---|---|
| Alopecia | Alopecia may lead to the loss of hair on the head and extra portions of the person's body | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |
| Obesity | Obesity causes diabetes to worsen faster | Diabetes → 1 | 10<br>11<br>01<br>00 | Nondiabetes → 0 | 00<br>01<br>10<br>11 |

based on the original input, colour fundus, and black–white image colours. The original input fundus images are formulated as∈P; P = {p/p∈{color←1, black-white←0}}, and the trained model predicts probability scores corresponding to∀q for an input p based on p∈{color←1, black-white←0}, which are treated as the expected output. In step 10, the image pre-processing techniques are implemented using dual-image multilayer ResNet algorithmic sequences by resizing the fundus-colored and black-white images by 312 × 312×3 data matrices for each image and integrating various data augmentation techniques such as zooming →z, trimming →t, rotary motion →rm, horizontal spin →hs, vertical spin → vs, breadth move → bm and height move → hm. The various trial cases are implemented based on training sampling with a neutral value of 1.0 in steach ep.11, and Max_features can signify the number of landscapes to be reflected for the best split; possible values are auto, sqrt, log2, and default value = None in step 12.

Every specific augmentation technique used in the image pre-processing, including trimming (t), rotary motion (rm), horizontal spin (hs), vertical spin (vs), breadth move (bm), and height move (hm), as well as zooming (z), had an impact on the overall robustness of the model. For example, by simulating various zoom levels, the model was able to recognize features at varying scales. This is especially crucial for FSC-CFI fundus and SSC-BWI black-white images, since the degree of magnification can change how lesions or other abnormalities appear. Because of its robustness, the model performs better when applied to a variety of imaging devices and protocols that may record images at different scales; The trimming caused by concentrating on various areas of an image, which aids in the model's ability to recognize important details even when they are partially hidden or near the margins, because of its robustness, the model is ready for real-world situations in which the field of view may be restricted or in which artifacts may cause portions of the image to become obscured. Additionally, because rotating images affect the rotary motion, the model becomes invariant to the orientation of the FSC-CFI fundus and SSC-BWI black-white images. This is essential because, depending on the imaging arrangement, pictures might be taken at various angles or orientations. Because of its robustness, the model is better able to identify images accurately regardless of their orientation, which is especially useful in clinical contexts where consistency isn't always ensured. More varied training examples were produced via the spin and move operations, which altered the location and rotation of the images. Spin and move procedures can mimic changes in the eye's location that occur naturally during imaging. Because of its resilience, the model can be trained to continue operating even in situations when image alignment or centering isn't exact, as it frequently is in real-world scenarios.

By exposing the model to a wider range of variations in handling real-world variations in FSC-CFI fundus and SSC-BWI black-white images, the model has analyzed the impact of augmentations on its performance. This has helped the model learn to generalize better to unseen data, which is crucial in medical imaging because patient variability can result in significant differences in image appearance. It primarily focused on image-related complexity, including performance on noisy data and prevention of overfitting. In pre-processing experimental stage, the augmentations help mitigate overfitting by creating a more diverse training dataset, encouraging the model to focus on relevant features rather than memorizing specific instances. Images of the FSC-CFI fundus and SSC-BWI black-white images taken in real life could have artifacts, noise, or inconsistent data. Through training the model to be resistant to these flaws, augmentations increase the model's dependability in real-world scenarios.

In step 13, the DL-based inception-ResNet multilayer classification-stage (CS) and regression-stage (RS) methods are applied for future stages, as follows:

Blended Hybrid Model (BHM) ←{{CS1, CS2}, RS}

Generated two imlr1 and imlr2 with distinct momentums, such as Mt1 and Mt2, for

Every∀bhm∈BHM

Every imlr = 0.001 to 0.005 and momentum = 0.4 to 0.8

Every epoch=1 to 360, do Both tiny clusters (Pmini, Qmini) ∈ (P, Q) then processed test conditions from TC1 to TC2

Then processed SGD based optimization test conditions from TC1 to TC5 for DM classificationwith the help of various epoch and Mt values.

Then processed the Dual-Image Multilayer Mapping Image-based Regression Model (RM) andconducted two TCs through Adam with tuning parameters for regression-based DM.

In addition to the previous step, the features considered from $\forall\tau\eta\varepsilon$ bhm∈ BHM are fed to an MLPclassifier to generate the dual image-based multilayer mapping approach in step 14. Finally, the various test cases are framed according to steps 7 to 14, the implications of which are shown in the algorithmic sequences.

## IV. EXPERIMENTAL SETUP

All the machine learning models were applied to the diabetic risk prediction dataset, and the outcomes are shown

as the F scores in Table 3. According to the results and evaluation of the model, the maximum classification accuracy is achieved by the 2DCNN, and the accuracy of the dataset-measured classification is 99%. A performance comparison with the F score and chi-squared test results is shown in Table 2 for the best categorical input and categorical output. The best models for predicting diabetes in the dataset are the XGBoost, random forest, and decision tree models.

The training proficiency of the proposed methodology depends on balanced, errorless data, so it must be tuned for training and testing purposes. The images are processed using the suggested CNN-implicated dual-image multilayer mapping technique based on DL. The black-white image-based data and color fundus, including Non-DR, MiDR, MoDR, SeDR, and PrDR, are evenly split depending on each DR stage, assisting the model in removing any disparities throughout the suggested approach's training and testing. Following equal sizing, all of the input color fundus and black-and-white photographs are processed; each image is adjusted in a methodical sequence, and the combination images are chosen for further analysis of the grading and prediction-making.

### A. DISPARITY PARAMETERS

The various dependable to non-dependable disparity parameters are important for the early detection of DR stages based on impact-causing symptoms. This section categorized and analysed various D-NonDMs based on their variations and grouped them into two clusters, as shown in Table 2. The first cluster framed the data values based on true, true-positive (TP) and true-negative (TN) information. True is indicated by '1', true-positive is indicated by '11', and true-negative is indicated by '10'. The second cluster frames the data values based on false, false-positive (FP) and false-negative (FN) results. True is indicated by '0', False-Positive by 01, and False-Negative by 00.

The hybrid Inception-ResNet (HI-RN)-implicated multilayer mapping-based transfer learning method has been used for automated prediction classification and regression environments to obtain accurate five-stage DR models, namely, Non-DR, MiDR, MoDR, SeDR, and PrDR, based on {000, 001, 010, 011, 100}. The cluster class 1 (CC-1) occurrences are formulated with 10, 11, 01, and 00, whereas the cluster class 2 (CC-2) occurrences developed with 00, 01, 10, and 11 are supported by analysis of the DR stages to determine the impact range of the DR.

## V. RESULTS AND DISCUSSION

The multi-decision inception-ResNet blended hybrid model includes many layers of dual image-based constraints to handle both sequential and non-sequential pictures. A multi-layered transfer learning process with 172 weighted layers is used to train the suggested model. 86 of these weighted levels are plotted in relation to color fundus images and the remaining 86 weighted layers are connected to black-and-white images. Individual image layer mapping and decision-making (DM) are performed through an integrated mapping technique with epoch-influenced values, as shown in the table. 3.

### A. A DATASET INTEGRATION

Providing comprehensive details about the division of the EyePACS and APTOS datasets, the specific training cum testing splits, filtering criteria, preprocessing steps, and final dataset statistics is essential for replicability and understanding the model's performance which will help to enhance the reliability of the findings and supports future research efforts in diabetic retinopathy detection.

The EyePACS dataset consists of fundus images used for diabetic retinopathy detection, typically labeled according to severity levels ranging from 0 which considered as no-DR to 4 which considered as proliferative-DR. It contains a diverse set of images representing various demographics and clinical conditions. The APTOS dataset consists of labeled fundus images for diabetic retinopathy detection. It includes images with a similar grading system but may vary in quality and acquisition methods.

#### 1) TRAINING, TESTING AND VALIDATION SETS AND SUBSETS

This section articulated divisions of datasets based on training, and testing sets and supporting subsets along with its validations. Often, a common approach is to use 70-80% of the data for training, 10-15% for validation, and the remaining 10-15% for testing. In sampling, it has ensured that each subset reflects the overall distribution of diabetic retinopathy grades, stratified sampling employed, which maintains the proportion of each class in all subsets.

#### 2) IMAGE SELECTION AND FILTERING

In training and testing, the specific criteria initiated for selecting images from the EyePACS and APTOS through main and sub-clusters based quality control and exclusion of anomalies. Images are controlled the filtering process based on quality metrics such as resolution, clarity, reflection-rate and lighting. In the basic stage of input, exclude anomalies such as artifacts, poor lighting, and low resolution issues and applied augmentation and normalization techniques to the images before training the data. In normalization adjusted pixel values to a standard range, typically (0, 1) to ensure consistency across images. In augmentation, the dual-image multilayer ResNet algorithmic sequences are used to implement the image pre-processing techniques. The fundus-colored and black-and-white images are resized by $312 \times 312 \times 3$ data matrices for each image, and various data augmentation techniques, including zooming (z), trimming (t), rotary motion (rm), horizontal spin (hs), vertical spin (vs), breadth move (bm), and height move (hm), are integrated. The augmentation techniques and methods described in section ''Model Architecture''.

**TABLE 3.** Individual image layer mapping and decision-making (DM) process through an integrated mapping technique with poch-influenced values.

| | | Mapping Technique | | | | Epoch Influenced Values | |
|---|---|---|---|---|---|---|---|
| | | Learning rate (ilmr) level | imlr value | Momentum (m) level | Momentum value | Epoch level | Epoch value |
| Individual Image Layer Mapping | (BHM) ←{{CM1, CM2}, RM} | bhm-imlr-1 | 0.001 | m1 | 0.4 | bhm-e1 | 1 |
| | | Bhm-imlr-2 | 0.005 | m2 | 0.8 | bhm-e2 | 360 |
| | | Bhm-durable value | 0.001 ➔0.005 | durable momentum (dumo) | dumo:{0.4 ➔0.8} | bhm-durable epoch (duep) | duep:{1 ➔350} |
| Decision Making (dm) | Stochastic Gradient Decent optimization (SGD) ← Parameters tuning and integration | dm-imlr1 | 0.0001 | dm1 | 0.4 | dm-e1 | epochs>70 |
| | | dm-imlr2 | 0.0002 | dm2 | 0.5 | dm-e2 | epochs>140 |
| | | dm-imlr3 | 0.0003 | dm3 | 0.6 | dm-e3 | epochs>210 |
| | | dm-imlr4 | 0.0004 | dm4 | 0.7 | dm-e4 | epochs>280 |
| | | dm-imlr5 | 0.0005 | dm5 | 0.8 | dm-e5 | epochs>350 |
| | | : | : | : | : | : | : |
| | | : | : | : | : | : | : |
| | | dm-imlr-n | n -value | i-value | ith value | dm-e ith value | epochs> ith value |

#### 3) DATASET SIZE, DIVERSITY AND FINAL DATASET COMPOSITION

In test cases, the total number of images used from each dataset and their distribution across different diabetic retinopathy severity levels. Data about the demographic diversity such as age, ethnicity of the images provided. The final composition of dataset initiated with balancing classes and ensured uniformity in preprocessing, and considered the number of training 70-80%, validation 10-15%, and testing 10-15% of images, as well as their corresponding labels and classes.

### B. SYMMETRIC ALLOCATION OF IMAGES

Initially, this equal allocation of images processed based on the clustering technique, the total sample images stores in one cluster, then further divided and processed with two sub-clusters, first sub-cluster (FSC-CFI) representing the color fundus images and same set of images converted and replicated into black-and-white images called as SSC-BWI, kept into second sub-cluster. Then the layering technique initiated from this stage. The first set of layers framed based on FSC-CFI and second set of layers framed based on SSC-BWI. The proposed methodology integrates 172 weighted layers, with 86 layers dedicated to processing color fundus images and the remaining 86 layers for grayscale images. This symmetric allocation of images helps to analyze the images and correcting features. Comparably, the first and second moments of the color image are framed based on exponential decomposition rates, representing cim1 and cim2, respectively, and the first and second moments of the black-and-white image are based on exponential decomposition rates, bwim1 and bwim2, respectively. From FSC-CFI, in first momentum 43 layers and in second momentum 43 layers are initiated out of 86 layers along with their corresponding weights. In same way from SSC-BWI first momentum 43 layers and in second momentum 43 layers are initiated out of 86 layers along with their corresponding weights.

The equal allocation of layers to both image types initiated based the rgb2gray function, which is based on parallel computing, transforms RGB true color images, also known as fundus images, into grayscale images, also known as black-and-white images. By removing the hue and saturation information from RGB images while keeping the luminance, the rgb2gray function turns them to grayscale. The color image, denoted by 3 numeric arrays as pxq, the color pictures of data types double and single are expected to have values in the range of 0 to 1 by the rgb2gray function. The grayscale image, or black-and-white image, was then returned as a pxq numeric array. The color map in grayscale, returned as a c-by-3 numeric matrix with values between 0 and 1. The unlabeled dataset is divided into k different clusters using the K-Means clustering technique (K-MC), which is applied to each set of sub-cluster images such as FSC-CFI and SSC-BWI. This ensures that each dataset belongs to only one group with similar properties. The FSC-CFI and SSC-BWI features were identified and analyzed using the K-MS to determine the severity of DRP. Following the identification of the characteristics, each sub-cluster image's internal behavior was examined. These common and unique features were then linked with individual FSC-CFI and SSC-BWI images.

This symmetric allocation of images helps to optimize the model performance in detecting DR by identifying the accurate features and its internal behavior factors such as color properties, size, shape, connectivity to neighbor dots, numbers of dots and so on. The various color properties counted for further process in DR, because color is often connected with the object or scene that is being shown in the image, it is the most frequently used visual feature in image retrieval. The color properties are more resilient to changes in the size, orientation, and viewing angle of the image than other visual qualities are. Equally, color features are highly responsive to changes in image brightness and grayscale information. Because diseased images need to be stained, color normalization is necessary before extracting color characteristics [56], [57], [58]. A color image is far

more vivid than a grayscale image because the original color carries crucial information. Color information should be included in color image hashing since color images with the color feature $f_{p,q}^{FSC-CFI}$ are currently far more common than grayscale photos such color feature $f_{p,q}^{SSC-BWI}$ as shown in the equations (1) to equation (4).

$$\int_{p,q}^{FSC\text{-}CFI} = \frac{\sum_{p=1}^{n}\sum_{q=1}^{n} Rxy + Gxy + Bxy}{3} \tag{1}$$

$$\int_{p,q}^{SSC\text{-}BWI} = \frac{\sum_{p=1}^{n}\sum_{q=1}^{n} BLACKxy + WHITExy}{2} \tag{2}$$

Accumulated cluster properties of FSC-CFI

$$\begin{aligned} FSC-CFI(P,Q) = & \int_{1,1}^{FSC-CFI} dpq \quad \int_{1,2}^{FSC-CFI} dpq \ldots\ldots \int_{1,p/q}^{FSC-CFI} dpq \\ & \int_{2,1}^{FSC-CFI} dpq \quad \int_{2,1}^{FSC-CFI} dpq \int_{2,p/q}^{FSC-CFI} dpq \\ & \int_{p/q,1}^{FSC-CFI} dpq \quad \int_{p/q,1}^{FSC-CFI} dpq \int_{p/q,p/q}^{FSC-CFI} dpq \end{aligned} \tag{3}$$

Accumulated cluster properties of SSC-BWI

$$\begin{aligned} SSC-BWI(P,Q) = & \int_{1,1}^{FSC-BWI} dpq \quad \int_{1,2}^{FSC-BWI} dpq \ldots\ldots \int_{1,p/q}^{FSC-BWI} dpq \\ & \int_{2,1}^{FSC-BWI} dpq \quad \int_{2,1}^{FSC-BWI} dpq \int_{2,p/q}^{FSC-BWI} dpq \\ & \int_{p/q,1}^{FSC-BWI} dpq \quad \int_{p/q,1}^{FSC-BWI} dpq \int_{p/q,p/q}^{FSC-BWI} dpq \end{aligned} \tag{4}$$

Five assertions addressing the impact of a 172-layer multi-decision Inception-ResNet-blended hybrid model's complexity on computing efficiency and deployment feasibility in real-world contexts are presented in the experimental scenario and test cases. These include: increased computational demand by the vast design of a 172-layer model needs large computational resources, leading to longer training durations and higher energy consumption, which might impair efficiency, especially in contexts with limited computational capability; The intricacy of the model requires significant memory for both training and inference of the FSC-CFI fundus and SSC-BWI black-white images. Latency in real-time application based images due to the depth and complexity of the model may introduce latency during inference, which can be problematic for real-time applications like medical imaging diagnostics, where timely decision-making is critical.

This may hinder the model's ability to be deployed on edge devices or in low-resource environments with limited memory capacity. As a result, significant efforts in model optimization techniques, like pruning, quantization, or using model distillation, may be necessary to improve deployment feasibility and reduce complexity while maintaining performance, enabling broader usability in real-world scenarios. A few scalability issues may arise when the proposed model is deployed across multiple devices using FSC-CFI fundus and SSC-BWI black-white images, or in a distributed computing environment, where each instance needs sufficient computational resources to meet performance standards.

### C. TRANSFER LEARNING PROCESS

The transfer learning process for the 172-layer structure likely involved careful selection of a pre-trained model, such as a hybrid of Inception and ResNet, followed by adjustments to training protocols, including feature extraction or fine-tuning, learning rate scheduling, and data augmentation. It also involved the leveraging pre-trained models to improve the performance of a new model on a specific task, in this case, the detection of diabetic retinopathy using a 172-layer structure with both color and black-and-white images. This approach enables the model to leverage existing knowledge from large datasets while adapting to the specific nuances of the diabetic retinopathy detection task using both color and black-and-white images. The detailed explanation of how this process may have been tuned and which pre-trained models might have been used:

#### 1) SELECTION OF PRE-TRAINED MODELS

##### a: COMMON CHOICES

Pre-trained models like Inception, ResNet, EfficientNet are popular due to their demonstrated performance on large datasets such as ImageNet. These models have been trained on vast amounts of diverse image data, enabling them to learn rich feature representations that can be fine-tuned for specific tasks.

##### b: INCEPTION-ResNet HYBRID

For the 172-layer structure, a hybrid model combining features from both Inception and ResNet architectures may have been chosen. This combination benefits from the deep residual learning of ResNet, which helps mitigate the vanishing gradient problem, and the multi-path architecture of Inception, which allows the model to capture a variety of features at different scales.

#### 2) TUNING THE TRANSFER LEARNING PROCESS

Feature Extraction vs. Fine-Tuning: Depending on the task's complexity, the process can either involve using the pre-trained model as a fixed feature extractor by freezing or fine-tuning by unfreezing some layers to adapt the model to the specific dataset.

##### a: FEATURE EXTRACTION

In this approach, the convolutional base of the pre-trained model is used to extract features from the input images, and a new classifier (often a few dense layers) is added on top to make predictions.

**TABLE 4. Implications of the dual-image-based multilayer mapping approach with color images – FCI and black-white images – FBW-based momentum, inference and epoch range values.**

| Dual-image-based multilayer mapping approach | color image - FCI | | black–white image - FBW | |
|---|---|---|---|---|
| Momentum | First | second | first | second |
| Inference | cim1 ← AOpt | cim2 ← AOpt | bwim1← AOpt | Bwim2← AOpt |
| cim1 | 0.7 | 0.890 | 0.7 | 0.890 |
| cim2 | 0.7 | 0.890 | 0.7 | 0.890 |
| epoch range (er1) | 1 to 70 | 71 to 90 | 1 to 70 | 71 to 90 |
| epoch range (er2) | 71 to 90 | 91 to 140 | 71 to 90 | 91 to 140 |
| epoch range (er3) | 91 to 140 | 141 to 210 | 91 to 140 | 141 to 210 |
| epoch range (er4) | 141 to 210 | 211 to 270 | 141 to 210 | 211 to 270 |
| epoch range (er5) | 211 to 270 | 271 to 310 | 211 to 270 | 271 to 310 |
| epoch range (er6) | 271 to 310 | 311 to 360 | 271 to 310 | 311 to 360 |
| epoch range (er7) | 311 to 360 | >360 | 311 to 360 | >360 |

### *b: FINE-TUNING*

In this approach, after training the new classifier, some of the upper layers of the pre-trained model can be unfrozen and fine-tuned with a smaller learning rate. This helps the model adjust its weights slightly based on the specific characteristics of the new dataset.

### 3) TRAINING PROCESS ADJUSTMENTS

Learning Rate Scheduling: A learning rate schedule may be implemented to gradually decrease the learning rate during training. This helps in stabilizing training and allows for more refined weight updates in later epochs.

Data Augmentation: To increase the diversity of the training data, various data augmentation techniques such as rotation, zoom, and flips can be applied, which helps prevent overfitting and improves generalization.

Batch Normalization and Dropout: These techniques may be used within the model to stabilize training and reduce overfitting by regularizing the model.

### 4) INPUT DATA PREPARATION

Color and Black-and-White Images: The model may have been tuned to handle both color and grayscale images by ensuring that the input layer can accommodate different channel configurations. This could involve preprocessing techniques that normalize or adjust the input dimensions and channel depths accordingly.

### 5) EVALUATION AND VALIDATION

Cross-Validation: The tuning process likely included cross-validation techniques to evaluate model performance across different subsets of the dataset, ensuring that the model generalizes well and is not biased towards a specific subset.

**TABLE 5. Comparison between machine learning algorithms with F scores and chi-square tests.**

| Machine Learning Algorithm | F1 Score | Chi-square test |
|---|---|---|
| Naive Bayes | 85.57% | 85.59% |
| Logistic Regression | 89.42% | 88.46% |
| Decision Tree | 96.15% | 96.15% |
| Random Forest | 96.15 | 97.11% |
| SVM | 95.19% | 93.26% |
| K-Nearest neighbor | 91.34% | 89.42% |
| XGBoost | 96.15% | 97.11% |
| Voting classifier | 94.23% | 94.23% |
| Proposed methodology | 98.1% | 97.8% |

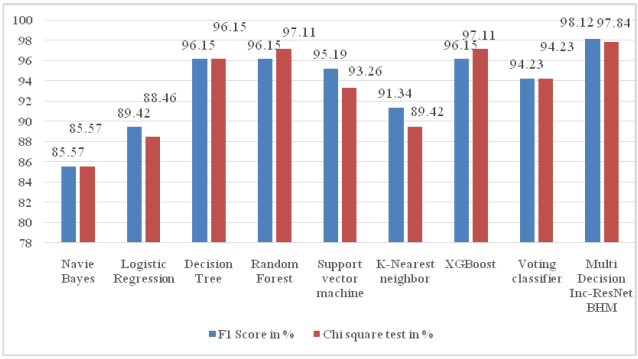



**FIGURE 5. Comparison between machine learning algorithms with F scores and chi-square tests.**

The implications of the dual-image-based multilayer mapping approach with color images, FCIs, and black-white images, FBW-based momentum, and inference with epoch range values are shown in Table 4.

Table 5 and Figure 5 present comparisons between machine learning algorithms with F scores and chi-square tests. The color fundus and black-white image-based data were plotted through 86+86 interconnections and interrelationships for analysing DR stages. It is pre-processed with various training phases and supporting sub testing steps. Every phase out of 13 comprises eight layers of convolutions with dependable, non-dependable data matrices, which are integrated with the number of repetitions of internal connections by composing global and specific features. The sample test cases are prepared for the F1 score by integrating the chi-square test mechanism.

## VI. CONCLUSION

This research proposed several layers of dual image-based parameters integrated by the multi-decision inception-ResNet blended hybrid model to analyze both sequential and non-sequential pictures. The multilayer-based transfer learning technique used to train the suggested model is tuned with 172 weighted multi-layers. It is plotted with 86 weighted layers linked to color fundus images and an additional 86 weighted layers linked to black-and-white images. It is

pre-processed with various training phases and supporting sub-testing stages. Every stage of the thirteen steps comprises eight layers of convolutions with dependable, non-dependable data matrices, which are integrated with the number of repetitions of internal connections by composing global and specific features. Sample test cases are prepared for the finite score by combining the chi-square test mechanism. Future research could include testing diverse datasets and adding new data features to refine the model's accuracy compared with the other models.

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## AUTHOR'S CONTRIBUTIONS

Santosh Kumar Henge was primarily responsible for developing the model architecture and conducting experiments, along with significant writing and organizational input. Nikhil Reddy Viraati focused on data collection, preprocessing, and model validation. Musaed Alhussein contributed to the theoretical framework and initial model design. Ajay Shriram Kushwaha, the corresponding author, played a key role in reviewing and refining the manuscript for correctness and coordinating the contributions of all authors. Khursheed Aurangzeb was involved in statistical analysis and methodology drafting, while Ravleen Singh contributed to coding, model implementation, and preparation of figures and tables.

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## COMPETING INTERESTS

The authors declare that they have no competing interests.

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